Agent-Based Semiology

Simulating office occupation patterns with conversation-based social models

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The importance of fostering formal and informal conversation to optimize office space performance has been well researched since the introduction of the 1970s cybernetic office layout strategies and recent research suggests, that formal and informal conversations at work can no longer be meaningfully separated, making efficient conversation patterns even more central to a successful office layout in the age of knowledge economy. In such a setup, social factors, like hierarchy, group membership, or expertise, contribute more to the formation of an office's spatial occupation patterns than the space's morphology itself. Consequently, standard tools of space evaluation, such as Space Syntax, that rely on the analysis of a space's topological description, yield inconclusive results, as the quantitative description of the space can no longer be matched to the changing patterns of interactions observed in that space. The research methodology described in this paper, therefore, aims to optimize contemporary office environments in a different way. Embedded in the conceptual framework of agent-based simulation, this research does not foreground the configuration of space itself but focuses on developing a population of agents sophisticated enough to allow for the emergence of an a simplified, yet plausibly life-like collective office scenario. Here, special occupation patterns evolve over time based on series of subsequent communication events between all agents in a space, where participants, locations, total numbers of various types of conversations, and durations depend on previous events as well as on a simplified social model. Different office scenarios are then analyzed against a set of selected criteria to identify successful office configurations. This paper describes the methodology's underlying concepts and setup, introduces the agent-based simulations that were developed and presents and speculate about the preliminary research results and findings.

Keywords: *Design Methodology, Agent-based Modelling, Office Space Simulation.*

INTRODUCTION

The importance of fostering formal and informal conversation to optimize office space performance has been a well-researched topic since the introduction of the 1970s cybernetic office layout strategies.

While in traditional Taylorist office spaces conversation, i.e., the exchange of information, was - conceptually detached from the production process, thus leaving scientific management solely concerned with the optimization of the flow of material and goods, at some point the strictly linear organization of work could no longer successfully sustain the increasing complexity of office cooperation.

Consequently, innovative office design proposals, most notably German Quickborner team's Bürolandschaft and Probst and Nelson's Action Office were developed, which were able to map the more intricate and networked patterns of interaction the gradual introduction of immaterial labor brought about (Saval, 2014).

Most recently knowledge economy has become the most important factor the developed world's service sector. Work processes are entirely based on the continuous flow of information and knowledge. Contemporary office layout, therefore, have become more informal and hybrid, as their performance hinges on their capability to facilitate communication within an office population in various formal and informal ways.

As office space layouts have become more complex and evolved, so have methods of office space evaluation and optimization.

ANALYSIS OF OFFICE SPACE TYPOLOGIES

For a long time, the understanding of office typology was derived from the Taylorist factory floor layout and the assessment and optimization of its manual labor processes which were relying on a linear transfer of work and goods. (Myerson and Ross, 2006).

The performance of a Taylorist workspace could be easily evaluated, for example by counting the units produced or the casefiles completed.

Innovations in office space design were brought about by two main factors: the steady raise in workflow complexity and the shift from manual towards immaterial work. These changes in the workflows underlying organizational principles have resulted in more complex office space environments and, consequently, resulted in the development of more advanced tools to evaluate their performance.

Analysis of modernist office spaces

Modernist office space concepts such as Quickborner's Bürolandschaft or the Action Office still use scientific management's strictly linear understanding of workflow (Rumpfhuber, 2019), however, their differentiated layouts are the result of optimization processes with the ambition to foster formal and informal communication flow, which was considered essential in their cybernetic organization model (Kockelkorn. 2008). The performance of such multi-directional spaces with their intricate patterns of interaction have proven more difficult to assess.

Today's most successful tool for office space research is Bill Hillier's Space Syntax, developed in the 1970s as one of the first coherent theories about two-dimensional spatial morphologies and the social logics that determine their use. Proposing to analyze the built environment "[…] at the level of [a] system of spatial relations" (Hillier and Hanson, 2003: 3) Hillier and his team developed a set of analytical tools to understand between space and its societal implications.

In this type of research, the morphological attributes of a space and the interaction patterns of its population are put into relation with each other by analyzing its geometrical properties with Space Syntax' tools and subsequently correlating the results with information about the social interactions in it, normally taken from surveys done by the users of that space (see for example: Peponis et al., 2007).

However, even this successful research methodology, however, appears to have its limits.

Space syntax analysis of knowledge economy spaces

Space Syntax relies in its methodologies on the geometric description of the inhabited spaces it aims to analyze (Bafna, 2003). Knowledge work driven contemporary office occupation patterns, on the other hand, mostly rely on the dynamic interactions of an office space population rather than on its physical boundaries.

As research shows, in these cases social factors within an office ecology, such as expertise, hierarchy, departmental affiliation, or personal relationships increasingly outweigh the implications of office space morphology (Grajevsky, 1993; Peponis et al., 2007; Markhede and Koch, 2007).

In knowledge space analysis, therefore, the quantitative description of the spatial layout can no longer be clearly correlated to the interaction patterns observed in that space; space syntax analysis leads to inconclusive results (Steen and Markhede, 2010; Groves and Marlow, 2016a).

OFFICE SPACE SIMULATION WITH AGENT-BASED MODELS

It is the hypothesis of this research, that patterns of interaction and communication in knowledge work office environment are no longer static in nature but show emergent behavior, which needs to be understood as the result of countless small individual interactions by the office population over time, which gradually add up to the complex and unpredictable state of the overall system.

The results of such bottom-up processes can no longer be analyzed by examining geometric configurations but need to be understood by exploring the intricate relationships between the actors in that space.

They can, therefore, be simulated using agent-based modelling (ABM), which is in its most concise definition "a computational method that enables a researcher [to] experiment with models composed of agents that interact within an environment." (Gilbert, 2008: 2).

As a computational method, they are computer programs which are developed to understand the social phenomena that occur in the real world (Sawyer, 2004).

First introduced to computer science in 1987, when Craig Reynolds (1987) successfully computationally modelled the flocking behavior of birds in a simulation called "Boids", its field of application has grown considerably, now including numerous areas of research, including social sciences.

For all areas of research, numerous agent-based simulations have been developed, all with special scientific focus, various levels of complexity and each catering to different segments of the scientific community (Abar, Theodoropoulos, Lemarinier and O'Hare, 2017).

For the research of social phenomena, NetLogo is at this point the most popular software tool (Railsback et al., 2006).

Building on its tradition as an educational tool, it is particularly widespread within the academic community, suitable for researchers without explicit computational background and allows to build on a digital archive of "[…] a large number of agent-based models written in NetLogo in a wide variety of domains" (Wilensky and Rand, 2015: xiv).

For these reasons, agent-based simulations for this research will also be programmed in NetLogo. It is fast, scalable and data extraction is easy. However, NetLogo's representational capacities are very basic and visual output is limited to simple 2.5D graphic representation.

DEVELOPING SOCIAL MODELS FOR AGENT-BASED SIMULATIONS

Architecture has only recently adopted agent-based models as a methodology for spatial simulation and optimization when engineers started to model dynamic crowd movements such as building evacuation patterns or pedestrian simulations.

Like natural phenomena such as flocks of birds or schools of fish, these models only operationalize the spatial relations between agents, negotiating agent properties such as cohesion, distance, or alignment. Contemporary office occupation patterns, however, are the result of non-spatial factors such as other members of the agent population, or a space's semiological features.

Basic considerations

To account for this change in object-space relation, this research aims to develop a more complex model of agent behaviour, in which an agent's actions are no longer in response to a space's morphology, but by its social interactions with fellow agents and semiological or environmental features.

To this end, the agent-based simulation needs to incorporate a series of additional features that go

beyond the mere processing of spatial quantities, of which the two most crucial ones are agent differentiation and architectural frame dependency (Schumacher, 2019: 51).

First, agent need to be individually differentiated according to their social role within the office ecology as an agent's status, hierarchical position, knowledge, group affiliation or importance within the social network will result in behavioural differences upon agent interaction.

Also, agent need to show architectural frame dependency, allowing them to exhibit differentiated behavioural patterns depending on their location in the space and the semiological features in it.

The aim is to model a heterogenous population of agents with individual properties and a sophisticated set of rules which enables the agent-based simulation to "[…] move from the current evacuation- and traffic-engineering crowds to architectural and semiological crowds as the basis for generalized life-process simulation" (Schumacher, 2016: 112).

Simplification and bounded rationality

Social science research in any analytical symbol system (Ostrom, 1988) is always conducted by building simplified models of social phenomena. Agent-based social simulations need to be built as simplified models to be able to understand the artificial system better than the social system observed in the real world for one to simulate the other (Axtell and Epstein, 1994). Consequently, agents are always modelled with limited capabilities. They are modelled to act with bounded rationality (Simon, 1957).

Within the scope of this research the social capabilities of agents are gradually added to a set of basic properties such as environmental awareness, purposeful walks, collision detection, or object and agent recognition. The challenge is, therefore, "[...] not to limit the rationality of agents, but to extend their intelligence to the point where they could make decisions of the same sophistication as is commonplace among people" (Gilbert, 2008: 16).

But, observing a gradual build-up of agent complexity with correspondingly improving simulation outcomes, how can one make decisions about the minimum level of complexity required for satisfactory simulation results? Miller and Page (2007) suggest turning this investigative process around and reducing agent sophistication step by step from full sophistication. The level of minimally required agent complexity can then be pinpointed to wherever a small change in agent sophistication leads to critical changes in the overall system's behavior.

It is worth noting that overly complex models are not successful, either, as there is a trade-off between analytical tractability and descriptive accuracy (Fagiolo, Windrum and Moneta, 2006: 10).

The importance of conversation in the contemporary workplace

The Taylorist model of office organization which was mainly concerned with the efficient distribution of goods and materials had little need for information interchange, as planning and production were separate processes. As the nature of work gradually shifted from physical transactions to the exchange of knowledge and skills, however, conversations have become a crucial element of contemporary workflow.

The introduction of Cybernetics and its idea of information flow and feedback loops into office design in the late 1960s led to the design of flexible networks of information interchange that were conceived to allow for the emergence of formal and informal communication (Kockelkorn, 2007: 6).

As of today, knowledge economy represents approximately one third of all economic activities in Western European countries (Eurostat, 2013: 115) and services constitute more than 73% of all EU employment (Eurostat, 2021: 43). This focus on immaterial work with its emphasis on ideas and knowledge and need to organize its flow efficiently has seen the emergence of entirely new types of working patterns of movement and collaboration (Greene and Myerson, 2011).

In this context, Rumpfhuber (2019) points out, that once office production is mainly immaterial in nature, there can be no longer any production without communication.

Marazzi (2007) shows that in contemporary working processes communication and production always overlap and, therefore, communication gains directly productive value.

Berardi (2009) argues that the constant emotional and affective investment of knowledge workers in their work starts to eradicate the difference between private and professional interactions making all interaction equally important for office space performance.

It is worth noting, that personal interaction strengthens digital relationships and vice versa (Groves and Marlow, 2016b: 160) and that despite recent developments towards home office work the desire for personal office interaction remains unchanged (PwC et al., 2013).

Towards a conversation-based social model

First, a basic behavioural model needs to be established that allows for flawless navigation through physical simulation space. Features of this basic model will need to include boundary detection, agent movement and orientation, agent perception, object and agent recognition, and obstacle detection and avoidance.

In a second step, a social model needs to be implemented in which the agents' social behavioural patterns are embedded. Within the scope of this research, this model will focus on establishing interaction patterns based on conversational logic depending on the agent's individual characteristics.

Agents are grouped in two teams. After entry, each agent will continuously decide to either engage in individual agent interaction with a member of the other team or approach the nearest high table and interact with a group of agents. Every agent will talk to every agent of the other team only once and visit every high table once. If an agent has decided on agent interaction but no agent is available, it will approach the nearest available table instead, likewise if an agent has decided on table interaction and there is no table available, it will look for a conversation partner instead. After each task completed an agent will navigate the space in search of the next available agent or table. In random intervals, agents receive a call on their mobile phone. In that case they interrupt their pending actions and make their way towards the far end of the space for extended privacy to resume their action once the call is finished. Once an agent has talked to all the available conversation partners and visited all the tables, it will be idly, but still available to other agents for agent interaction.

Simulation setup

Agent-based simulation are designed to share the same set of characteristic features: Ontological correspondence, a representation of the environment in question, a set of heterogeneous agents, and agent interactions based on bounded rationality (Fagiolo, Windrum, and Moneta, 2006).

For a first simulation to test the system's capability to emulate simple social behavior, as a simulation environment, we define an open office space with the size of 40 by 20 meters with two opposite entrances and a fixed number of high tables as furniture elements which will act as semiological features in the space.

Agents are divided in two equal groups who enter from opposite sides of the room. Upon entry each agent's behavioral pattern is determined by its social interaction parameters with every agent keeping track of its individual progress, thus ensuring heterogeneity.

Two office layouts will be tested in this research. In the first layout (scenario 1), all six tables are arranged in a tight array at the far end of the simulation space. In the second layout (scenario 2), six tables are distributed randomly across the space. In each scenario every agent will, as described before, talk to all agents of the other group once while visiting tables and receiving phone calls.

Basic Agent Attributes		
Parameter	Value	Basis for Value
Body Dimensions	Agent Body Circle Diameter = 0.5 m	Based on average shoulder width; Dreyfuss (1967)
Speed Distribution	Standard Speed = 0.83 m/s	Average speed calculated based on data from Rimea (2016)
	Standard Step Length = 65 cm	
	Standard Deviation = 0.15	Standard Deviation Based on Fruin (1987)
Maximum Turn Rate	45.0	Based on qualitative model observations.
Cone of Vision	Length = 45 m	Length is set to make sure the entire space is visible to the agent.
	Angle = 120.0	Set for peripheral vision, Yarbus (1967).
Proxemics	Intimate Space Radius = 0.8 m	Concept and radii based on Hall (1966), values for Intimate Space and Personal Space slightly adjusted based on previous quantitative study model
	Personal Space Radius = 1.2 m	
	Social Space Radius = 2.0 m	

Table 1
Basic agent attributes

Simulation Setup Values	
Parameter	Defaut Value
Objects	Number of High Tables = 6
Agents	Number of Agents Team A = 12
	Number of Agents Team B = 12
Probabilities	Team A to select Action 3 = 0.7
	Team A to select Action 4 = 0.3
	Team B to select Action 3 = 0.7
	Team B to select Action 4 = 0.3
Interaction Parameters	Max. Number of Agent per Table = 3
	Average Agent Interaction Length = 90 sec.
	Average Object Interaction Length = 120 sec.
	Average Phone Call Length = 100 sec.
	Random Phone Call Intervall = 120 sec.

Table 2
Simulation setup values and social agent attributes

The simulation ends automatically as soon as all agents have fulfilled their assigned social tasks. The hypothesis, then, is, that by comparing the simulation runtime of both scenarios we can identify the more performative office layout, as it allows for the faster handling of a fixed number of social interactions.

Basic agent attributes are informed by research on the basic principles and fundamental physical parameters of human action, such as anthropometrics, human performance metrics, human perception, and proxemics (see table 1).

Social agent attributes for this generic test are - in lack of a corresponding real-world scenario - set to reasonable numbers or subjected to randomization but can always be replaced with empirical information later (see table 2). As we are aiming for a comparative study to assess different office space furniture layouts, their absolute values are not crucial to the simulation result, however, they need to remain the same across all simulation setups.

Each of the two simulations will be run 10 times with an identical setup to check for consistency and variation and identify potential outliers. During runtime, relevant data such as agent positions, locations of interactions, or numbers of conversations are recorded, stored ready for export at the end of the simulation (see figure 1).

VALIDATING ACCURACY AND REALISM

Validating accuracy and realism of an agent-based simulation is by no means an easy task. While other, more traditional social research methodologies have developed a canon of accepted methods of empirical validation, evidence-based assessment is often considered a key weakness in agent-based social simulations.

Fagiolo, Windrum and Moneta (2006: 6) identify agent-based modelling's focus on simulating qualitative changes rather than on quantitative changes as the key issue. Also, as agent-based modelling is generally used to investigate emergent and non-linear social phenomena, quite often, empirical data is not readily available for comparison.

To assess an agent-based model's performance, therefore, Axtell and Epstein (1994) propose a multi-level evaluation system based on qualitative and quantitative analysis rating models from merely appearing to be in visual agreement with reality (level 0) to exhibiting quantitative and qualitative agreement on both, the system's micro (the individual agent) and macro (the system's overall behaviour) level (level 3).

To check the simulation for qualitative plausibility, the simulation's visual output is analyzed during runtime on the level of the system's overall appearance. Upon inspection the agent population appears to exhibit differentiated and life-like behaviour in purposeful navigation, agent-to-agent interaction, and agent-to-object interaction.

Assessing the simulation on the agent level it can be monitored that agents show individual and distinctive behaviours towards agents with other social properties as well towards objects distributed

in space and that selection processes of social interaction follow a complex set of rules and do not result from simple rules such as distance or visibility.

This observation is confirmed by comparing simulations' the resulting heatmaps (which depict the agent positions over time superimposed in one single image thus rendering more frequently used patches darker than others): As the objects' positions vary in different simulation setups so do the individual agents' cumulated locations in space over time.

Examining the quantitative performance shows that for each setup simulated simulation length is similar (with small differences which seems reasonable when simulating emergent behaviour) for all 10 simulation runs, whereas length notably differs from one setup to the other. This is an indicator for the simulation's consistency across different simulated scenarios.

ASSESSING THE PERFORMANCE OF OFFICE LAYOUS

Once an agent-based model is found to be sufficiently robust and reliable in its qualitative and quantitative performance, no absolutely accurate values are necessary to identify the best design alternative. Schumacher (2020: 76) argues, that "[t]he presumption that relative performance advantages have veracity even if absolute performance measures are false seems plausible enough to merit investing in the simulation methodology" which at some point will lead to necessary empirical calibration and validation.

Upon examination, the two series of simulation show different results for performance value and spatial distribution.

In simulation scenario 1 with the table array at the far end of the space agents need an average of 4.665 iteration to complete their social tasks, with 4.202 iterations being the shortest and 5.257 the longest (compare figure 2). In simulations scenario 2 with a random table distribution, agents finish after an average of 3.817 iterations with 3.488 being the shortest and 4.063 the longest (compare figure 3).

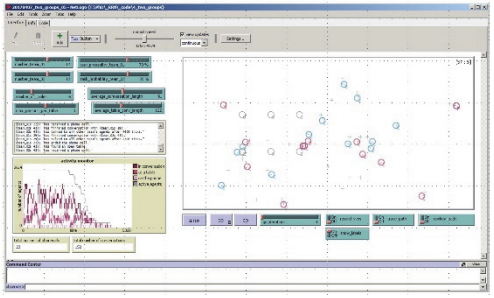

Figure 1
NetLogo's graphic interface during simulation

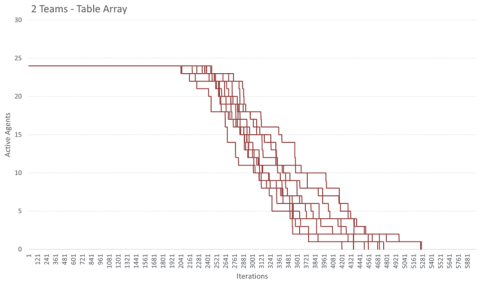

Figure 2
Performance data of scenario 1

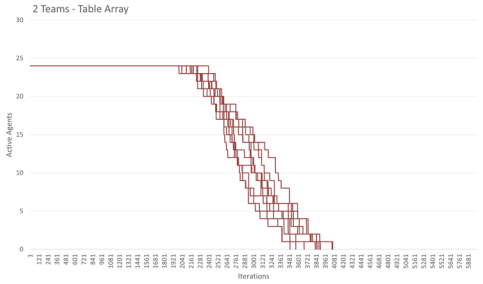

Figure 3
Performance data of scenario 2

The results suggest that scenario 2 has a relative performance advantage over scenario 1 which can be attributed to the fact that a central communication space loosely surrounded by tables allows for more efficient patterns of agent interaction and table navigation than a lateral separation of tables and interaction space.

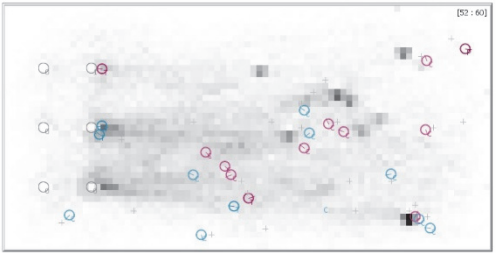

Figure 4
Heatmap for scenario 1

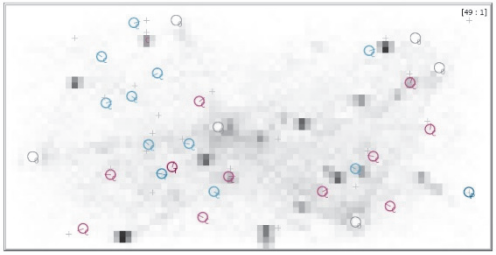

Figure 5
Heatmap for scenario 2

Again, the heatmaps seem to confirm this speculation: they show patterns of use changing according to different layouts with more frequently used areas of space shifting towards the middle in case of scenario 2. Comparing the heatmaps also confirms reduced overall use of space for scenario 2, suggesting shorter paths of circulation throughout the simulation (compare figure 4 and 5).

CONCLUSIONS

When driven by simple conversation-based social models that govern individual agent-to-agent and agent-to-object interaction, agents clearly show diverging interaction sequences and differentiated patterns of space occupation solely based on different furniture layout. Agent populations show the capacity to complete their social task with varying efficiency depending on changing object configurations in the space, thus allowing to identify more performative office furnishing solutions.

At this point it is impossible to assess the simulation's realism in respect to the absolute performance values collected during the simulation. Some space layouts, however, show relative performance advantages over others, indicating clear potential for further investigations. Design rules generalized from the findings of this research can also be made operational in future research when approaching similar design challenges.

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REFERENCES

Abar S., Theodoropoulos G., Lemarinier, P. and O'Hare, G. (2017) 'Agent Based Modelling and Simulation tools: A review of the state-of-art software' Computer Science Review 24, pp. 13-33.

Axtell, R. and Epstein, J. (1994). 'Agent Based Modelling: Understanding Our Creations' in The Bulletin of the Santa Fe Institute, Winter 1994, pp. 28–32.

Bafna, S. (2003). 'Space Syntax A Brief Introduction to Its Logic and Analytical Techniques' Environment and Behaviour. 35, pp. 17-29.

Berardi, F. (2009). The Sould at Work. From Alienation to Autonomy. South Pasadena: Semiotext(e).

Dreyfuss, H. (1967). The Measure of Man - Human Factors in Design. New York: Whitney Library of Design.

Eurostat (ed.). (2013). Science, Technology, and Innovation in Europe. Luxembourg: Publication Office of the European Union.

Eurostat (ed.). (2021). Key figures on Europe. 2021 Edition. Luxembourg: Publication Office of the European Union.

Fagiolo, G., Windrum, P., and Moneta, A. (2006). 'Empirical validation of Agent-Based Models: A Critical Survey' in LEM Working Paper Series No. 2006/14. Pisa: Sant' Anna School of Advanced

Studies, Laboratory of Economics and Management.
Fruin, J. (1987). Pedestrian Planning and Design. Revised Edition. Alabama: Elevator World Inc.
Gilbert, N. (2008). Agent-Based Models. Thousand Oaks: Sage Publications.
Grajevski, T. (1993). 'The SAS Head Office Space - Spatial Configuration and Interaction Patterns'. Nordic Journal of Architectural Research, Vol. 2. pp. 63-74.
Greene, C. and Myerson, J. (2011) 'Space for Thought: Designing for Knowledge Workers' Facilities, Vol. 29 Issue: 1/2, pp. 19-30.
Groves, K. and Marlow, O. (2016a). 'Interview with Kerstin Sailer' in Spaces for Innovation - The Design and Science of Inspiring Environments. Amsterdam: Frame Publishers, pp. 61-64.
Groves, K. and Marlow, O. (2016b). 'The Spatial Context. Location, Proximity, and Interaction' in Spaces for Innovation - The Design and Science of Inspiring Environments. Amsterdam: Frame Publishers, pp. 157-164.
Hall, E. (1966). The Hidden Dimension. Anchor Books, New York: Random House.
Hillier, B. and Hanson, J. (2003) The Social Logic of Space. Cambridge: Cambridge University Press.
Kockelkorn, A. (2008). ‚Bürolandschaft – eine vergessene Reformstrategie der deutschen Nachkriegsmoderne' in ARCH+ Zeitschrift für Architektur und Städtebau, Vol. 186/187, pp 6-7.
Marazzi, C. (2007). Capital and Affects. The Politics of the Language Economy. Los Angeles: Semiotext(e).
Markhede, H., Koch, D. (2007). 'Positioning Analysis: Social Structures in Configurative Modelling' in Proceedings of the 6th International Space Syntax Symposium. Istanbul. pp. 069-1-069-14.
Miller, J. and Page, S. (2007). Complex Adaptive Systems. An Introduction to Computational Models of Social Life. Princeton: Princeton University Press.
Myerson, J. and Ross, P. (2006). Space to Work. New Office Design. London: Laurence King Publishing.
Ostrom, T. (1988). 'Computer simulation: the third symbol system' in Journal of Experimental Social Psychology, 24, pp. 381–392.
Peponis, J., Bafna, S. et al. (2007). 'Designing Space to Support Knowledge Work' Environment and Behaviour 39. pp. 815 – 840.
PwC, University of Southern California, and London Business School (eds.). (2013). NextGen: A Global Generational Study.
Railsback, S., Lytinen, S. and Jackson, S. (2006). 'Agent-based Simulation Platforms: Review and Development Recommendations' SIMULATION. 82, Issue 9. pp. 609-623. DOI 10.1177/0037549706073695.
Reynolds, C. (1987). 'Flocks, herds, and schools: A distributed behavioural model' Proceedings of the 14th Annual Conference on Computer Graphics and Interactive Techniques ACM. 21 (4), pp. 25–34.
Rimea e.V. (2016). Richtlinie für Mikroskopische Entfluchtungsanalysen. Version 3.0. https://rimeaweb.files.wordpress.com/2016/06/rimea_richtlinie_3-0-0_-_d-e.pdf
Rumpfhuber, A. (2019). 'Labour'. AA Files No 76, 2019, pp 111-116.
Saval, N. (2014). Cubed – A Secret History of the Workplace. New York: Anchor Books.
Sawyer, R. (2004). 'Social explanation and computational simulation' Philosophical Explorations, 7(3), pp. 219-231.
Schumacher, P. (2016). 'Advanced Social Functionality Via Agent-Based Parametric Semiology' in Schumacher, P. (ed.) Parametricism 2.0. AD 02/2016, London: Wiley, pp. 108-113.
Schumacher, P. (2019) 'Digital' AA Files No 76, 2019, pp. 47-52.
Schumacher, P. (2020). 'From Intuition to Simulation" in Ozvaldic, M., Pap, B., Umbrasaite, I. (eds.) Positions: Unfolding Architectural Endeavours. Basel: Birkhäuser Verlag, pp. 69-76.
Simon, H. (1957). 'A behavioural model of rational choice' in Simon, H. (ed.) in Models of Man. New York: Wiley.

Steen, J. and Markhede, H. (2010). 'Spatial and Social Configurations in Offices' The Journal of Space Syntax 1(1). pp. 121–132.
Wilensky, U. and Rand, W. (2015). An Introduction to Agent-based Modelling: Modelling Natural, Social, and Engineered Complex Systems with NetLogo. Cambridge: MIT Press.
Yarbus, A. (1967). Eye Movements and Vision. New York: Springer Science and Business New York.